EKREM TURAN FIRAT

Undergraduate Researcher | Ankara / Sivas, Türkiye

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I am a third-year Electrical and Electronics Engineering student and an undergraduate researcher at METU Power Lab. My main research and work area is power electronics and the design of advanced hardware for this field.

Experience

METU Power Lab — *February 2025 – Ongoing*
Undergraduate Researcher — *ODTÜ, Ankara, Türkiye*

- *Designing, building and testing power electronics hardware, mostly TI C2000-controlled motor drivers with low phase inductance for TÜBİTAK and industry projects: schematic, PCB layout, assembly and bench tests.*
- *Developing and maintaining the lab's KiCad libraries and a Claude-based KiCad design assistant.*

AGH Power Electronics Laboratory — *October 2025 – February 2026*
Undergraduate Researcher (Erasmus+) — *AGH, Kraków, Poland*

- *Designed a GaN-based synchronous buck converter demo board: electrical design, component selection and Altium schematic.*

IEEE SCÜ Student Branch — *November 2024 – September 2025*
Committee Chairman — *Sivas, Türkiye*

- *Chair of the PELS (Power Electronics Society) committee.*
- *Worked on the technical and organizing team of international symposiums.*

Robolink Teknoloji — *June 2023 – August 2023*
Electronics Design Intern — *OSTİM, Ankara, Türkiye*

- *Designed an ESP32-based laser CNC motherboard on a 4-layer PCB in Altium CircuitStudio, with passive power protection, USB-to-TTL and stepper motor driver circuits.*
- *Assembled and tested the boards: SMD soldering, oscilloscope measurements and board bring-up.*

Robolink Teknoloji — *March 2022 – August 2022*
Electronics Design Intern — *OSTİM, Ankara, Türkiye*

- *Designed an independent stepper motor controller on a 2-layer PCB in EAGLE, which generates the pulse signal the driver needs on its own. It went into series production and was sold.*
- *Designed an ESP32 development board to verify the company's in-house design processes; it was manufactured and tested.*

Education

Sivas Cumhuriyet University — *September 2024 – Ongoing*
Bachelor of Science, Electrical & Electronics Engineering — *Sivas, Türkiye*
Current coursework: *Electrical Machines · Electric Power Generation, Transmission and Distribution*

AGH University of Krakow — *October 2025 – March 2026*
Erasmus+ Exchange, Electrical Engineering — *Kraków, Poland*
Relevant coursework: *3D Design with SolidWorks · Computer Measurement Systems (NI LabVIEW) · Introduction to AI*

Cezeri Green Technology Technical High School — *September 2020 – June 2024*
Technical Diploma, Electrical & Electronics Technology — *Ankara, Türkiye*

Projects

METU Power Lab — *ODTÜ, Ankara, Türkiye*
PCB Motor with Internal Driver — *October 2026 – Ongoing*

- *An axial-flux motor whose stator windings are traces on a 12-layer PCB, with the whole motor driver on a bay of the same board: three TI LMG2100 GaN half-bridges at 100 kHz (8.7 A rms, 15.4 A peak per phase), an STM32G491 MCU, linear and index Hall sensors for the absolute rotor angle, CAN FD and an isolated USB-C debugger.*
- *12 V input with reverse-polarity and hot-swap protection, plus a USB-C PD input. The rev A board (75 × 107.5 mm, 221 parts on both sides) is fully routed with a clean DRC.*

PowerLab KiCad Assistant — *October 2026 – Ongoing*

- *A KiCad 10 side panel that sets the rules for Claude in PCB design, so it can build an entire project to the METU Power Lab standard: requirements, schematic, libraries, placement, routing, ERC/DRC, design review and fabrication files.*

PL Driver Cycloid Board Design — *July 2025 – January 2026*

- *Embedded motor driver for a PCB motor with a cycloidal gear (TÜBİTAK 1001). I worked on the hardware design of this 250 W, low-phase-inductance, TI C2000 series DSP-controlled miniature driver.*

PL Driver Evaluation Board Design — *February 2025 – July 2025*

- *A 500 W (up to 2 kW) motor driver on an advanced 8-layer PCB with extensive communication interfaces (TÜBİTAK 1001). I worked on its hardware design, assembly and testing.*

AGH Power Electronics Laboratory — *AGH, Kraków, Poland*

GaN-Based High-Frequency Switched DC-DC Synchronous Buck Converter — *October 2025 – January 2026*

- *An educational demo board for laboratory classes, where students watch how the parts heat up as frequency, load and voltage change. I did the electrical design, component selection and Altium schematic; the project stopped at the design stage when my exchange ended.*

Robolink Teknoloji — *OSTİM, Ankara, Türkiye*

ESP32 Based Laser CNC Motherboard — *June 2023 – August 2023*

- *Designed 4-layer PCBs for ESP32-based laser CNCs running GRBL; the boards were manufactured and tested.*

ESP32 Development Board — *August 2022*

- *Designed a 2-layer ESP32 development board for testing ESP32-based applications within the company; it was manufactured and tested.*

Stepper Motor Driver — *March 2022 – July 2022*

- *Designed a 2-layer stepper motor controller that needs no software or external controller. It went into series production and was sold.*

Scholarships & Awards

TÜBİTAK — *April 2025 – November 2026*

1001 Program Undergraduate Researcher Scholarship

Memberships & Organizations

IEEE Power Electronics Society (PELS) — *November 2024 – Ongoing*

Student Member

Skills

PCB Design	*Altium, KiCad, EAGLE, Proteus, CircuitStudio*
3D Design	*Fusion 360, SolidWorks*
Simulation	*LTspice*
Embedded	*Basic embedded software for microcontrollers*
Lab	*Oscilloscope and lab measurement equipment, SMD soldering, board assembly and testing*

Languages

Turkish	*Mother Tongue*
English	*B2*
Polish	*A1 (Beginner)*